



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Information Technology **Academic Year 2025 - 2026 (Even Semester)**

Semester, Degree & Branch: II Semester, B.Tech & IT

Course Code & Title: IT25201& Foundations of Data Science using Python

Name of the Faculty member (s): Mrs. P.Ramya, AP/IT

Innovative Practice Description

Unit / Topic: Module VI / Visualizing time series Data-Line plots to show trends over time

Course Outcome: CO6

Topic Learning Outcome: -

Activity Chosen: Learning by Teaching

Justification:

In the study of Data Visualization, Time Series analysis using Line Plots is a critical component of both theoretical and laboratory curricula. Through the Learning by Teaching approach, students not only gain technical proficiency in using libraries like Matplotlib or Pandas to plot temporal data but also enhance their conceptual understanding by explaining trends, seasonality, and noise to their peers. This peer-teaching strategy forces students to move beyond mere coding to master data storytelling, encouraging the collaboration and communication skills necessary to translate complex numerical fluctuations into clear, actionable insights.

Time Allotted for the Activity: 20 Minutes

Details of the Implementation:

- Ms. G. Subathra Devi was entrusted with the topic "Visualizing Time Series Data with Line Plots," having been given a week's lead time for preparation.
- She developed the instructional materials and led the session.
- Her presentation spanned roughly 20 minutes, providing a comprehensive theoretical foundation followed by a detailed coding explanation.
- The delivery included a live demonstration to bridge the gap between concept and implementation.
- The execution of this "Learning by Teaching" initiative is documented in Figure 1.

CO – PO / PSO mapping:

CO	PO1	PO2	PO9	PO10	PO11	PSO1
CO6	3	3	2	3	2	3

(1 – Low 2 – Moderate 3 – High)

PO/PSO Mapped:

Innovative Practice	PO1	PO2	PO9	PO10	PO11	PSO1
	3	3	2	3	2	3
Justification for Correlation	Apply mathematical and programming principles to interpret temporal data trends and fluctuations.	Identify and analyze patterns like seasonality and noise in complex datasets to reach meaningful conclusions.	Demonstrate leadership and teaching capabilities by guiding peers through technical coding workflows.	Effectively communicate complex data insights and technical logic through clear visual storytelling and verbal explanation.	Adapt to evolving data visualization libraries and tools essential for continuous professional development.	Ability to design and implement efficient Python-based solutions for analyzing and visualizing time-stamped datasets.

- **Images / Screenshot of the practice:**



Figure 1. G. Subathra Devi- "Visualizing Time Series Data with Line Plots"

Reflective Critique:

- ❖ **Feedback of practice from students and other stakeholders:**

- Participants noted that the session clarified the complexities of temporal data and increased their comfort level in using Python for trend analysis.
- The requirement to lead the class drove the presenter to explore advanced plotting techniques, resulting in a more robust understanding of data patterns and anomalies.
- Faculty observed that the student-led format sparked dynamic discussions, making the

classroom environment more collaborative than conventional teaching.

- The approach was praised by stakeholders as a creative way to turn passive learning into an active, hands-on mastery of data visualization.

❖ ***Benefit of the practice:***

- Learners gained a sophisticated perspective on linear trends and seasonality, learning how to extract insights from sequential data points.
- The exercise sharpened their programming agility with time-stamped datasets and strengthened their ability to communicate findings through clear, professional graphs.
- It successfully aligned classroom theory with real-world application, fulfilling the objectives of experiential and learner-centric education.

❖ ***Challenges faced in implementation:***

- Several students struggled early on with the syntax for date indexing and the logic behind smoothing out noisy data.
- Although the activity was allotted 20 minutes, the extensive coding demonstration and Q&A session pushed the total time to 30 minutes.

References:

<https://www.ritrjpm.ac.in/images/B.Tech-IT/2025-2026/5.Learning%20by%20teaching.pdf>

Signature of Faculty Member

HOD